

ZAGREB PLESO, Croatia

WMO: 142410

Lat: 45.7292N

Lon: 16.0539E

Elev: 109

StdP: 100.02

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.8	-8.0	-13.7	1.2	-9.0	-10.9	1.5	-6.0	10.0	1.1	8.9	2.1	1.6	250	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	33.0	22.0	31.2	21.4	29.9	21.0	23.2	30.4	22.4	29.3	21.5	28.1	3.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.0	15.8	26.7	20.0	14.9	25.9	19.2	14.1	24.7	69.3	30.5	66.1	29.5	63.3	28.2	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
8.5	7.2	6.0	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-14.9	35.3	3.7	2.2	-17.6	36.9	-19.8	38.2	-21.9	39.4	-24.6	41.0
			-15.2	24.8	3.7	1.1	-17.8	25.5	-20.0	26.1	-22.1	26.7	-24.8	27.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.7	0.8	2.7	7.2	11.9	16.4	20.3	22.0	21.6	16.8	11.9	6.7	1.5	(6)
(7)		DBStd	8.46	4.55	4.76	4.35	3.65	3.38	3.37	2.90	3.03	3.35	4.10	4.46	4.48	(7)
(8)		HDD10.0	1031	286	207	106	23	1	0	0	0	28	115	265		(8)
(9)		HDD18.3	2769	544	438	346	192	78	19	5	7	69	201	348	522	(9)
(10)		CDD10.0	1651	1	2	19	81	201	308	371	361	203	87	17	1	(10)
(11)		CDD18.3	347	0	0	0	1	19	78	117	109	22	2	0	0	(11)
(12)		CDH23.3	3703	0	0	1	31	247	827	1236	1126	217	20	0	0	(12)
(13)		CDH26.7	1329	0	0	0	3	48	286	492	452	48	1	0	0	(13)
(14)	Wind	WSAvg	2.1	2.0	2.3	2.8	2.6	2.5	2.2	2.1	1.9	1.9	1.8	1.9	1.9	(14)
(15)	Precipitation	PrecAvg	877	52	49	55	64	77	95	83	89	91	80	85	61	(15)
(16)		PrecMax	1235	174	130	126	157	189	194	225	260	236	224	261	146	(16)
(17)		PrecMin	521	4	3	4	4	10	37	26	1	22	0	1	0	(17)
(18)		PrecStd	157	34	30	26	33	39	32	42	57	49	53	51	34	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.8	17.2	22.2	26.6	30.1	33.8	35.6	36.1	30.8	25.8	20.9	15.2	(19)
MCWB			10.7	10.3	12.6	16.5	19.8	22.5	21.7	21.0	19.6	17.9	14.9	11.2	(20)	
2%		DB	12.0	14.8	19.3	23.9	28.0	31.8	33.0	33.1	28.0	23.2	17.8	12.3	(21)	
		MCWB	8.8	9.4	11.5	14.8	18.7	22.0	22.2	21.9	19.7	17.3	13.5	9.5	(22)	
5%		DB	9.8	12.2	17.1	21.9	26.1	30.0	31.1	31.1	26.0	21.0	15.2	10.0	(23)	
		MCWB	7.8	8.2	10.8	14.1	17.8	21.2	21.6	21.1	18.7	16.3	12.2	8.2	(24)	
10%		DB	7.2	10.0	14.9	19.8	24.1	28.1	29.6	29.1	23.9	18.7	13.2	7.8	(25)	
		MCWB	5.7	6.9	9.7	12.9	16.8	20.2	21.0	20.5	17.8	15.0	11.2	6.8	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.0	11.2	13.4	17.4	21.1	24.1	24.0	24.2	21.4	19.1	15.2	11.7	(27)
MCDWB			13.9	16.0	19.9	24.6	27.7	31.1	31.1	31.3	27.4	24.0	19.1	14.3	(28)	
2%		WB	9.3	9.8	12.2	15.6	19.6	22.9	23.1	23.0	20.2	17.7	13.9	9.9	(29)	
		MCDWB	11.7	14.1	18.1	22.1	26.4	29.8	30.4	30.2	26.3	22.1	17.0	12.1	(30)	
5%		WB	7.7	8.5	11.1	14.5	18.4	21.8	22.4	22.0	19.2	16.6	12.7	8.2	(31)	
		MCDWB	9.5	11.9	16.2	20.3	24.8	28.4	29.4	28.7	24.9	20.2	15.2	9.9	(32)	
10%		WB	5.9	7.2	10.0	13.4	17.4	20.7	21.5	21.1	18.3	15.4	11.2	6.6	(33)	
		MCDWB	7.0	9.6	14.3	18.5	23.2	26.9	28.0	27.6	22.9	18.3	13.4	7.5	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.9	9.2	10.9	11.6	11.8	12.2	12.4	11.0	10.1	7.6	6.6	(35)	
MCDBR			11.0	13.6	15.3	16.2	15.3	14.7	15.2	15.9	14.4	13.3	11.7	10.5	(36)	
MCWBR			8.0	8.9	8.6	8.4	7.4	6.4	5.9	6.1	6.8	7.7	7.8	7.9	(37)	
5% WB		MCDWR	10.2	12.4	13.3	14.1	13.9	13.8	13.6	13.9	13.2	12.1	10.8	9.8	(38)	
		MCWBR	7.6	8.3	7.6	7.6	7.1	6.3	6.0	6.1	6.6	7.4	7.4	7.5	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.337	0.363	0.399	0.448	0.436	0.440	0.439	0.440	0.426	0.398	0.364	0.336	(40)	
(41)		taud	2.347	2.287	2.228	2.111	2.195	2.229	2.229	2.239	2.246	2.307	2.351	2.383	(41)	
(42)		Ebn at Noon	751	800	821	810	835	832	827	810	784	749	709	709	(42)	
(43)		Edh at Noon	78	99	121	148	141	137	135	129	118	97	77	68	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.23	1.98	3.18	4.37	5.41	6.09	6.13	5.39	3.72	2.43	1.35	1.00	(44)	
(45)		RadStd	0.19	0.34	0.44	0.50	0.50	0.44	0.40	0.59	0.48	0.28	0.18	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.52	N/A	N/A	+1.03	N/A	N/A	N/A	N/A	+97	+62	(46)
(47)	Regional (50 neighbors)	+0.46	N/A	N/A	+0.82	+0.45	+0.38	N/A	-117	+87	+56	(47)

Nomenclature: See separate page